

AAPG2020	ArtISMo		PRCE
Coordinated by:	Ali ZEMOUCHE	48 months	580 092,10 €
Scientific evaluation committee CE48			

Intelligent Estimation Algorithms for Smart Mobility

Partner	Name	First name	Current position	Role & responsibilities in the project	p.m
CRAN	ZEMOUCHE	Ali	Associate Prof.	Coordinator of the project Manager of WP1 and WP3 Major contribution/expertise to all WPs	24
CRAN	ALMA	Marouane	Associate Prof.	Adaptive observer, instrumentation WP3, WP4, WP5	12
CRAN	DELATTRE	Cédric	Associate Prof.	Stability analysis, network communication WP3, WP5	12
CRAN	RAFARALAHY	Hugues	Associate Prof.	Modeling – Estimation WP2, WP3, WP5	14.4
CRAN	BOUTAT-BADDAS	Latifa	Associate Prof.	Control design and Estimation WP3, WP4	12
CRAN	ZASADZINSKI	Michel	Professor	Stochastic systems, Prediction/Estimation WP3, WP4, WP5	14.4
IBISC	ICHALAL	Dalil	Associate Prof.	IBISC's scientific leader Manager of WP4 Modeling – Estimation: WP2, WP3, WP5	15
IBISC	MAMMAR	Saïd	Professor	Vehicle dynamics and Control WP2, WP4, WP5	15
IBISC	NOUVELIERE	Lydie	Associate Prof.	Vehicle dynamics and Estimation WP2, WP3, WP4	15
IBISC	LAHIANI	Khaled	Engineer/PhD	Cyber-physical system, Observers WP4, WP5	15
IRSEEM	AHMED ALI	Sofiane	Assistant Prof.	IRSEEM's scientific leader Manager of WP2 Modeling – Estimation: WP3, WP5	12
IRSEEM	KHEMMAR	Redouane	Assistant Prof.	Perception – Deep-learning, computer vision WP1, WP2, WP5	12
IRSEEM	CABANI	Adnane	Assistant Prof.	Multi agent system, V2X communication WP2, WP5	9.6
IRSEEM	DUPUIS	Yohan	R & D Responsible	Perception, smart mobility WP2, WP5	9.6
IRSEEM	EL HOUDA HAJRI	Rihab	Research Engineer	Mechatronics system – Estimation WP2, WP4, WP5	7.6
IRSEEM	QUANGTRUC	Dam	PhD student	Nonlinear observer, autonomous vehicle WP3, WP5	6
IRSEEM	LANGLOIS	Nicolas	Professor	Diesel Engine Control WP2, WP5	3
FAAR	TOOM	Randolph	Director of Innov.	FAAR's scientific leader Manager of WP5 Automotive applications: WP1, WP5	4
FAAR	SALVATORE	Enzo	Engineer	Automotive applications WP5	2.5
FAAR	NANG-TCHEE	Eric	Engineer	Automotive applications WP1, WP5	2.5
FAAR	DUTHOIT	Melchior	Engineer	Automotive applications WP1, WP5	2.5
FAAR	MARCHAND	Thomas	Engineer	Automotive applications WP1, WP5	2.5
LISEC	RAJAMANI	Rajesh	Professor	LISEC's scientific leader Expertise – advising on all WPs	4

Partner	Requested	Advisors	Position	Role & responsibilities in the project	p.m
CRAN	PhD1	CRAN/IRSEEM	PhD student	Work on the tasks of WP3 and partially on WP2 and WP5	36
IBISC	PhD2	IBISC/CRAN	PhD student	Work on the tasks of WP4 and partially on WP2 and WP5	36
CRAN	PostDoc1	CRAN	PostDoc	Work on the tasks of WP3 and WP5 (Collaboration with all partners)	18
IRSEEM	PostDoc2	IRSEEM	PostDoc	Work on the tasks of WP4 and WP5, and partially on WP2 (Collaboration with all partners)	18
CRAN	4 Internships	CRAN	MSc students	Work on WP3, WP5; connection to PhD1 and PostDoc1	24
IRSEEM	4 Internships	IRSEEM	MSc students	Work on WP2, WP5; connection to PhD1 and PostDoc2	24
IBISC	4 Internships	IBISC	MSc students	Work on WP4, WP5; connection to PhD2 and PostDoc2	24

Table 1: Summary table of persons involved in the project. (p.m): Involvement (person.month) throughout the project's total duration.

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Changes Compared to the Pre-proposal

Three members have been added in this full proposal to strengthen the consortium: Latifa BOUTAT-BADDAS from **CRAN**, Associate Professor; Khaled LAHIANI from **IBISC**, Engineer/PhD; Dam QUANG TRUC from **IRSEEM**, PhD student. Our industrial partner, **FAAR**, puts at the disposal of the project 4 engineers (E. Salvator, E. Nang-Tchee, M. Duthoit, T. Marchand) to work on the instrumentation and validation on real vehicles and with FAAR facilities.

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1 Proposal's Context, Positioning and Objectives

1.1 Objectives and research hypotheses

1.1.1 Context

The development of controllers with high performance and reliability for autonomous and connected vehicles will require real-time measurements or estimates of many variables on each vehicle [1]. Examples of variables that are needed for feedback include: longitudinal distances, velocities and accelerations of other nearby vehicles; lateral position of the vehicle in its own lane; vehicle yaw angle; slip angle; yaw rate; steering angle; lateral acceleration; and roll angle. There are also environmental variables which need to be measured such as tire-road friction coefficient, snow cover on road, and the presence of unexpected obstacles. Measurement of all of the above variables requires significant expense. Indeed, some of the sensors above, such as slip angle and roll angle, can be extremely expensive to measure, requiring sensors that cost thousands of dollars. For example the Datron optical sensor for measurement of slip angle has a price over 10k€. In addition, several variables cannot be measured due to unavailability of sensors (at any cost). Examples include positions, and accelerations of cars which are further upstream (*e.g.* lead car of a platoon). Only the position of the immediately preceding car ahead can currently be measured. Furthermore, autonomous and connected vehicle requires highly reliable sensors and actuators. Failure of any one sensor or actuator, due to faults, cyber-attacks or denial of service, can cause a disastrous accident. Hence reliable fault diagnostic and fault handling systems are also needed. Such systems cannot be based on hardware redundancy which requires many extra copies of the same sensors. Instead, they need to rely on estimation algorithms and analytical redundancy. For all the above reasons, the development of intelligent estimation algorithms is highly important for autonomous vehicles.

1.1.2 Objectives

Throughout this project we propose original ideas on estimation, which is a necessary and crucial step for reliability, resilience, and safety of connected and autonomous vehicles (CAVs). The overall objectives of the

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proposal consist in developing efficient estimation algorithms to reconstruct the unmeasurable state variables, which are required to design controllers and fault diagnostic schemes for CAVs. More specifically, the considered issues are safe and stable trajectory, estimation of faults in sensors and actuators, and cyber-attack detection. Below, we will describe the proposed ideas in detail.

1.1.2.a Learning-based neuro-adaptive observers for vehicle tracking

The idea we will explore in this project is the development and use of learning-based nonlinear observers. Several components on a vehicle (*e.g.* tires) have highly complex models whose parameters are difficult to obtain and also vary significantly with time. This proposal will therefore use a modeling approach consisting of a **combination of physically meaningful differential equations and adaptive online-learning-based neural networks to represent the vehicle dynamics**. In particular, well understood phenomena such as force balances, mechanical motion per Newton's laws, aerodynamic drag, rolling resistance, road grade, combined acceleration terms for lateral and roll accelerations and road bank angle influence will be modeled using analytical differential equations. Tire models for both lateral and longitudinal forces, the friction circle, engine maps, and suspension stiffness and damping characteristics will be modeled using neural networks whose weights can be initially obtained using training via back-propagation. In addition to initial training, model parameters for the neural networks and a subset of parameters for the physically meaningful differential equations will also be updated automatically online during regular vehicle use. The strategy is depicted in Figure 1(a), where y_1 represents measurements provided from radar reading and on-board sensors and y_2 are informations arising from wireless communication.

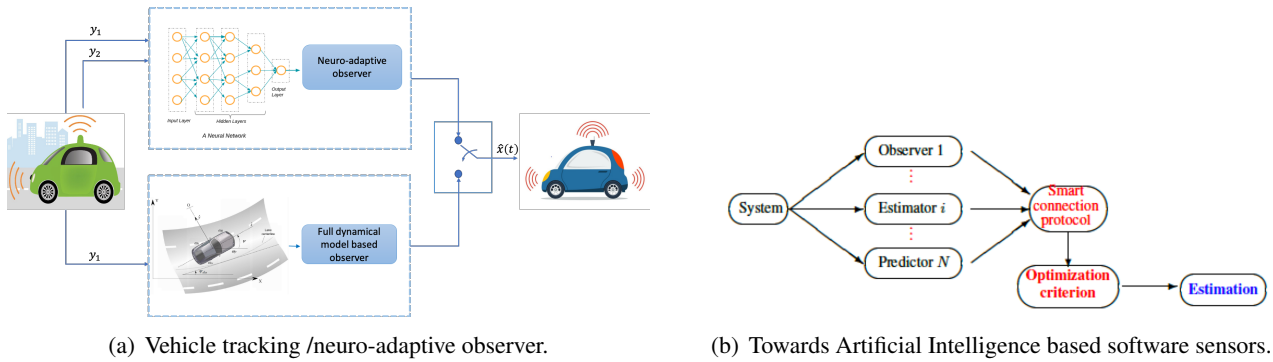


Figure 1: Intelligent estimation strategies for vehicle tracking.

The overall objective of this part is to develop the architecture for the combined modeling approach, to develop rigorous estimation algorithms for both parameter and state estimation, to enable dynamic changes in observer structure based on real-time vehicle maneuver and measurement availability and to validate the complete model and estimation system using data from CARSIM and from real vehicle experimental measurements provided by partners of the project. The above learning-based strategy requires significant new developments in estimation algorithms. It leads to several mathematical challenges that we describe in the developed methodology in Section 1.3.

1.1.2.b Towards intelligent estimation algorithms

Another objective of this proposal consists in proposing a **new and full system of software-enabled estimation algorithms to be included in many control strategies and diagnosis schemes to improve stability, reliability, and safety of connected vehicles**. To this end, as depicted in Figure 1(b), the aim is to combine the nice properties of statistical predictors [2], model-based methods [3], neuro-adaptive observers [4, 5], and learning-based techniques like the extremum seeking algorithm [6] to enhance the estimation and increase resilience, performances, and robustness of any control strategy related to connected vehicles.

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The challenge is to develop a smart connection protocol between the set of observers/estimators/predictors allowing the most accurate and reliable estimation. The combination of all these estimation techniques is worthily interesting for models with unknown inputs and/or unknown parameters. Among the problematics of autonomous and connected vehicles (in addition to the tracking problem mentioned above) where such intelligent estimation algorithms are crucial, we can cite the problem of cyber physical attacks or denial of service detection, disturbance and fault rejection, and data loss. Hence, estimating all the variables related to those factors may considerably increase resilience and reliability of interconnected vehicles. Due to the high sensitivity of autonomous vehicles to the above undesired factors, the development of smart estimation schemes is crucial and essential.

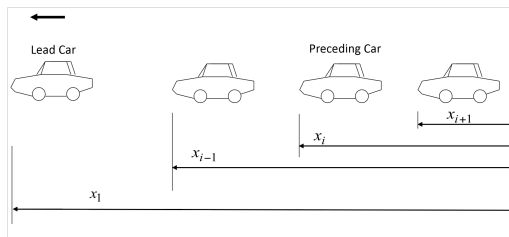
1.1.2.c Improving Resilience, Robustness, and Reliability of automated vehicles in a platoon

In the operation of interconnected vehicles, the network-based environment provides many advantages such as easy installation, maintenance with low cost, and high efficiency, so, combined with recent advanced communication technology, future controlled, connected vehicles are under network-based environment which also causes several undesired factors. These factors may be as follows:

- Disturbances and fault: faults of digital devices cause serious problems like degradation of system performance and even loss of human life. Therefore, the diagnosis of faults is needed by considering several engineering concepts like fault tolerant and fault diagnosis.
- Cyber-attacks: it is essential to develop some criteria for the vehicle systems to detect and counter cyber-attacks. We aim to consider several cyber-attack architectures involving nonlinear dynamics.
- Data loss: the influence of packet dropout will be strong when packet dropout and sampling are simultaneously occurred because, if packet dropout is occurred at a sampling instant, data will be missing during a sampling period. Therefore, we prepare some solutions for this complex situation. Specifically for this issue, the use and development of new statistical predictors will be of great interest.

Until now, many existing works solved the above stated problems individually. In this project, we will work on a unified solution counteracting all factors caused by the presence of network-based environments. We will investigate stability of the interconnected vehicle systems subject to those network-based factors. The objective is to develop a comprehensive theory from estimation viewpoint to improve resilience, robustness, and reliability of a large class of (inter)-connected nonlinear vehicle dynamics. The investigation of learning-based or data-driven based estimation algorithms and combining Unknown Input Observers (UIO) with nonlinear approximators will be raised in this proposal. Indeed, due to the smartness of new integrated technologies, the use of artificial intelligence-based algorithms may help to improve estimation and thus resilience and reliability of control systems in smart mobility. Combining analytical model-based techniques with learning-based algorithms [4] and statistical predictors [2] will enhance significantly the estimation quality.

Cars in a platoon (or cars in an arbitrary string of vehicles on a section of a highway) constitute interconnected dynamic systems, as depicted in Figure 2. The overall objective consists in developing complete **Resilient Robust and Reliable ($\mathbb{R}^3$)** control systems to face any problems in automated vehicles including vehicles operating in a platoon.



(a) Cars in a platoon.



(b) Automated Highway Demonstration in 1997 with Platoon Buick Le Saber Cars.

Figure 2: Automated vehicles operating in a platoon.

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If for any reason wireless communication links are lost or significantly disturbed with some cars (cyber-attacks, sensor faults, or other external factors), but not with other cars, then estimation can be done by using the partial wireless signals obtained and making up for the wireless signals missed, to get $\mathbb{R}^3$ —Controllers. This is especially possible because the radar signal of each car is also being transmitted by it wirelessly.

1.2 Position of the project as it relates to the state of the art

1.2.1 Originality and novelty of the proposal

The innovative character of the proposal lies in the development of general intelligent and sophisticated estimation algorithms compared to those existing in the literature for smart mobility. These algorithms will be specifically developed motivated by models issued from autonomous and connected vehicle problems. Developing a unified theory to face a large class of autonomous and connected vehicle issues related to estimation, by using fully nonlinear dynamics, is an original idea. **All the limitations of standard estimation algorithms will be overcome by developing learning based neuro-adaptive observers combined with approximators and statistical prediction approaches. This will lead to highly intelligent estimation algorithms.** The proposed ideas **will provide a complete and unique estimation system** covering a large class of estimation issues related to smart mobility.

1.2.2 Position of the proposal in relation to the state of the art

Among the known methods for the estimation of inter-vehicle variables in the field of autonomous and connected vehicles in general, we find the techniques based on well-known dynamic models (Slip angle, roll angle, etc), namely nonlinear observers based on LMIs, high-gain methodology, sliding mode, and extended Kalman filtering. These methods are efficient in good conditions, however, on the other hand, they fail in inconvenient situations (difficulty in obtaining a precise model, presence of external uncertainties and disturbances, measurement and actuator noise, etc). Although several methods have been developed to overcome the problem of disturbances and measurement and actuator noise, the availability of a precise dynamical model is necessary and constitutes a major drawback of model-based estimation methods. On the other hand, in the real-world applications, generally, we have a lot of data but no exact mathematical model for some components, or it is difficult to get a precise model in the presence of unexpected environmental conditions. Data-driven based methods have been used in the past for control and identification, however little attention has been paid to data-based nonlinear estimation.

◦ **State of the art on learning-observers:** More and more learning-based observers attract the attention of the automatic control community because of their benefits and their strength in the face of complex systems including new technologies, namely connected vehicles. Some early investigations into neuro-observers, for example, in [7], the authors assume model availability. However, the current wave of data-driven control has demonstrated the effectiveness of approximators in controlling systems in a model-free manner. Neuro-adaptive observers in the model-free setting were explored almost two decades ago in [5], where the authors proposed an adaptation rule for learning the weights of a linear-in-parameter neural network (LPNN) that results in uniformly ultimately bounded estimation error dynamics. Although this work has been adopted in multiple applications such as robot control, rotors, and more recently, wind turbines [8], the inherent assumptions and theory have hardly evolved. In most of these methodologies, the activation functions are considered to be radial basis functions, there is no measurement noise, and the theoretical guarantees of learning performance remain the same. A recent work has been proposed in [4] **by the scientific coordinator** (Ali ZEMOUCHE) of this proposal and **the partner leader** (Rajesh RAJAMANI), and their coworkers. Such a result has improved significantly the previous work in the literature on data-driven neuro-adaptive observers by using nonlinear activation functions, however, the topic still remains open until now. **This proposal aims to enhance the existing algorithms for data-based learning and apply it to intelligent transportation systems.**

◦ **State of the art on vehicle tracking:** Tracking of the motion of other vehicles on the road is an important problem that is frequently encountered in autonomous driving, as well as in collision avoidance and Adaptive

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Cruise Control (ACC) applications. There has been significant recent interest in the autonomous control of vehicles [9], but the topic of estimating the trajectories of other vehicles on a road in order to track them or avoid conflicts with them has not been sufficiently addressed. Collision avoidance and ACC systems typically use radar or laser sensors for measuring distances and azimuth angles to vehicles. With such sensors, radial distances and azimuth angles can be measured. In the case of vehicles in urban traffic, including vehicles at traffic intersections, the radar-measured variables are inadequate in order to fully predict the trajectories of vehicles. Both lateral and longitudinal distances and orientation of the vehicles are needed in order to accurately predict vehicle motion and provide appropriate warnings or automated driving actuation. Previous work in vehicle tracking has typically utilized interacting multiple model (IMM) filters [10]. The models used in the IMM filter typically include a “straight line driving” model and a “constant turn rate” model. Each model can be used for its respective driving scenario and is not applicable for the other scenario. To overcome the use of IMM, a new observer design has been proposed by W. JEON, A. ZEMOUCHE, and R. RAJAMANI in [11] by using an hybrid observer switching between different regions of monotonicity. Although the technique is efficient, however, the stability guarantee requires certain conservative dwell-time conditions, which depend on the vehicle motion. *We first aim to enhance the result and propose novel techniques, before combining with learning-based neuro-adaptive observers.* The consortium has a long experience in vehicle dynamics, estimation, stability, applied mathematics and optimization, to address these issues.

◦ **State of the art on cyber-attacks:** Potential cyber-attack threats on inter-vehicular communication systems and on sensor systems have been discussed previously in the literature [12]. False data injection (FDI) and Denial of Service (DoS) attacks can lead to significant problems in autonomous vehicles and connected vehicle systems that use inter-vehicular communication. An attacker can obtain access to the internal system of the vehicle or transmit false data to the control system. Also, sensor faults can be a serious source of problems for many intelligent transportation systems [13], [14]. Hence, it is necessary to have a resilient system for autonomous and/or connected vehicles that can be secure against cyber-attacks and sensor faults. Several other works in the literature have focused on cyber-attacks and sensor fault problems for connected vehicle systems. False data injection on acceleration information via V2V communication is considered and a Hidden Markov Model-based attack detection method is proposed for cooperative adaptive cruise control applications in [15]. Most of these works handle either cyber-attacks or sensor faults, but do not allow for both failures. This challenge motivated **the scientific coordinator** (Ali ZEMOUCHE) and **the partner leader** (Rajesh RAJAMANI), with their coworkers, to propose a unified estimation technique to handle simultaneously cyber-attack detection and sensor faults estimation [16]. Although the proposed algorithm has been evaluated by using experimental data from a real vehicle on the I-94 Highway in Minnesota (data obtained from sensors along an instrumented section of the I-94 westbound in Downtown Minneapolis), it concerns only linear systems. **In this project, we will develop a full nonlinear model-based estimators to address $\mathbb{R}^3$ -software sensors. Several cyber-attack architectures will be considered. Indeed, the estimation, in this case, uses data from wireless communication, which may be corrupted by malicious intruders.**

1.3 Methodology and risk management

We will structure the project for optimal management. A Gantt chart will be used to ensure daily management of human and material resources (*i.e.*, deadlines management, budget, intermediate targets and project review). The working plan we propose is based on a set of 5 work packages (WP), described below in detail. The global project provisional planning through Gantt chart is presented in Table 2.

• WP1: Project Management and Communication

Contributors—Deadlines—Objectives
Leader: Ali ZEMOUCHE (CRAN) Contributors: CRAN (6 p.m), IRSEEM (1 p.m), IBISC (1 p.m), FAAR (1 p.m) Deadlines (Start – End): T0 – T0+48 Objectives: This WP will be devoted to the methodology which will be followed by the partners to manage the project. The WP1 contains 3 tasks: T1.1–T1.2–T1.3.

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		Partners					1st Year: 2020-2021				2nd Year: 2021-2022				3rd Year: 2022-2023				4th Year: 2023-2024			
WP	Tasks	CRAN	IRSEEM	IBISC	LISEC	FAAR	T0+3	T0+6	T0+9	T0+12	T0+15	T0+18	T0+21	T0+24	T0+27	T0+30	T0+33	T0+36	T0+39	T0+42	T0+45	T0+48
WP1	T1.1																					
	T1.2																					
	T1.3																					
WP2	T2.1																					
	T2.2																					
	T2.3																					
WP3	T3.1																					
	T3.2																					
	T3.3																					
	T3.4																					
WP4	T4.1																					
	T4.2																					
	T4.3																					
WP5	T5.1																					
	T5.2																					
	T5.3																					
Deliverables/Deadlines																						
Partial Progress Reports																						
Consortium Arrangement																						
Project Management Review under Steering Committee																						
Final Report																						

Table 2: Project provisional planning with WPs (Gantt).

– *T1.1. Project Management:* The lead partner, in collaboration of the other partners, will use hybrid management-based on both V cycle and Agile management in order to improve the quality of project management. In this task, we will carry out different actions which represent a high quality tools for the project management: 1. Consortium agreement, 2. Gantt planning updated one time per week, 3. Website of the project updated continuously during the project at each time news are produced, 4. Risk analysis study, 5. Project management review at the end of each WP in order to validate the developments done.

– *T1.2. Communication and Results Dissemination:* Communication is very important in the success of our project. The lead partner will organise one first meeting as a Kick-Off meeting in order to present the project to partners through a provisional planning, budget, risk analysis, etc. The lead partner will organize on week meeting as a boot-camp per year and per partner in order to discuss about the progress of the project and the different contributions to develop by the consortium. Both administrative and scientific details will be studied in this annual meeting. In this WP, we will produce the whole of conferences that partners will make contribution to promote more the project. Review papers will also be presented and shared between the consortium partners. The lead partner, with the help of the other partners, will make available all intermediate progress report related to the project. We will also produce the final report to be submitted to the ANR office. The involved partners will provide the project with a cloud-based online platform for the collaborative dataset for smart mobility. The collaborative dataset will include data/images/videos and relevant informations.

– *Task 1.3. Open Innovation & Companies Transfer:* Finally, we will also engage with professional bodies and with the ANR team to identify other opportunities for exploitation by considering ways to transfer this innovation to companies collaborating with our organizations and specifically through our industrial partner, FAAR, willing that their potential customers that they are currently in contact with will integrate the outcome of this project for their products.

Deliverables–Risks and Fallback Solutions	
Deliverables	
– Website and internal communication tool	

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- Intermediate reports of progress at the end of every year, and final report
- Organization of meetings and a workshop
- Organization of professional meetings for interaction with companies

Risks and Fallback Solutions: In general, a risk can occur in the delivery deadline of the partners which could not be respected. However, this risk is very minor because some partners have already collaborated together (IRSEEM/FAAR, CRAN/LISEC, CRAN/IRSEEM) and then there are good relationships between them. As a fallback solution, if a moderate risk is occurred, is updating the Gantt chart and reorganization of resources between WPs. The role of master students will be useful for that. The work of PhD students and Postdocs will be closely monitored.

• **WP2: Vehicle Modeling, Environment Perception and Deep-Learning**

Contributors–Deadlines–Objectives
Leader: Sofiane AHMED ALI (IRSEEM) Contributors: CRAN (30p.m), IRSEEM (50.8p.m), IBISC (20p.m), FAAR (3p.m), LISEC (1p.m) Deadlines (Start – End): T0 – T0+45 Objectives: This WP will be devoted to benchmarking, modeling and environment perception in the goal to provide necessary informations for the development of estimation algorithms. First, we will construct a full nonlinear model which describes accurately the vehicle dynamic. Second, we will use computer vision and deep-learning to generate additional measurements for the observer design to enhance the estimation and to verify correctness of some measurements provided by LIDAR sensors. The WP2 contains 3 tasks: T2.1–T2.2–T2.3.

– T2.1. State of the Art and Benchmarking: In this task, the 3 partners CRAN, IRSEEM and IBISC will dedicate themselves to perform a state of the art on the tracking vehicle projects in the field of smart mobility. The goal of this task is to compare the different vehicle tracking projects currently implemented by the different car manufacturer. The next step is the benchmarking regarding existing solutions and projects. Subsequently, the sharing of knowledge bases in machine learning-based AI and neuro adaptive estimation algorithms for autonomous vehicles would be an essential step so that all partners can communicate their results.

– T2.2. Vehicle Modelling: In this task, the partners will dedicate themselves to develop a full nonlinear vehicle dynamics model upon which our intelligent estimation algorithms will be designed. This full nonlinear model will combine 3 different types of vehicle modelling approaches. The first model will describe the Kinematic and dynamic models for lateral vehicle dynamics. This model will be useful to ensure that the vehicle will stay in its center of line. The second model the partners concerns the vehicle longitudinal dynamics, which will be used for several applications such as radar-based collision avoidance systems, adaptive cruise control systems. The third model we aim to develop is an Power train model for either Gasoline, Diesel engines or hybrid vehicle. This model will be complementary to the previous longitudinal vehicle dynamic model, in order to enhance longitudinal control applications which will be carried out in the project. Finally the partners will put together these 3 models in one full nonlinear model which will be the base to achieve WP3 and WP4.

– T2.3. Environment Perception-based Deep-Learning: The main contribution in this task is to develop deep learning platform applied to road smart mobility, which will allow improving the quality of the estimation algorithms by generating additional output measurements. Indeed, observers need measurements, which could be collected from different sensors like RADAR, LIDAR, etc. This task is devoted to the environment perception-based computer vision and deep learning [17]. Usually, we measure distances and azimuth angles using radar and/or LIDAR sensors. The aims is to make new computer vision based measurements [18] as outputs in the observers to enhance estimation. Also, the computer vision data will serve to verify whether our measurements from radar are correctly associated with the correct cars, which lanes the cars are in, etc. In addition, computer vision can provide lateral position inside each lane. Deep-learning

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algorithms will also be used in the development of data-driven neuro-adaptive observers of WP3 and their generalization.

Deliverables–Risks and Fallback Solutions
Deliverables – Open-source library on the developed model in Matlab & Python, released on github – Open-source library on computer vision based dataset collection/camera RGB-D and radar measurements – Publications in journals (IEEE Sensors, IEEE TVT) and in conf. proceedings (ASME, IEEE) Risks and Fallback Solutions: The natural risk is the unavailability of dataset under which we will build and train our computer vision-based deep learning model. However, this risk vanishes because the consortium can cover this issue. Data can be generated by using vehicle platforms from FAAR, IRSEEM, and IBISC. Also our partner LISEC can collect enough data from sensors along an instrumented section of the I-94 westbound in Downtown Minneapolis.

• **WP3: Intelligent Software Sensors for Vehicle Tracking**

Contributors–Deadlines–Objectives
Leader: Ali ZEMOUCHE (CRAN) Partner contributors: CRAN (70p.m), IRSEEM (24p.m), IBISC (19p.m), LISEC (2p.m) Deadlines (Start – End): T0+6 – T0+45 Objectives: This work package is dedicated to development of novel intelligent software sensors for vehicle tracking problem. The overall objective is to introduce a new architecture of combined modeling approach consisting of a combination of physically meaningful differential equations and adaptive online-learning-based neural networks to represent vehicle dynamics. This work package will offer a basis of a new unified theory on nonlinear estimation for vehicle tracking, which will be used in the other work packages. The experience of the research team in the field of nonlinear estimation and autonomous vehicles allows to handle this WP3 without any significant risk issue. The WP3 contains 4 tasks: T3.1–T3.2–T3.3–T3.4.

– *T3.1. Regular Nonlinear Observer Design (RNOD):* Before tackling the strategy depicted in Figure 1(a), the development of a model-based observer for the dynamical model developed in WP1 should be investigated. Namely, we will work on nonlinear observer design for a model using nonlinear coupled tire force [1]. Usually the full nonlinear model is very large and complex, then we will study this full model and adapt it to our situation, namely vehicle tracking as presented in this project. Resilience and robustness of the observer, with respect to some time-varying parameters, will be investigated. LMI-based techniques with time-varying observer gains are one of the planned tracks. A general theory will be developed in this context. Also, as demonstrated in [11], the non-monotonicity of some nonlinearities affects feasibility of LMI conditions. The idea we plan to explore consists in introducing transformations to overcome the non-monotonicity and avoid infeasibility of the LMI-based design conditions. The use of a general hybrid observer to switch between several regions of monotonicity is also a solution.

– *T3.2. Neuro-Adaptive Nonlinear Software Sensors (NANS):* Data-driven neuro-adaptive observers are not developed enough to address the tracking problem described in previous sections. Indeed, the vehicle tracking problem will involve several parameters so that the available methods are unable to provide solutions. The model is under the form

$$\dot{x} = \varphi(x, u) \quad (1a)$$

$$y = Cx \quad (1b)$$

where the unknown nonlinearity $\varphi(x) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ represents the unmodeled dynamics of the system. Under some assumptions, we can invoke neural network approximability results to represent the unknown

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nonlinearity as

$$\varphi(x, u) = W\sigma(x, u) + \varepsilon(x, u), \quad (2)$$

where $W \in \mathbb{R}^{n \times N}$ is a constant but unknown matrix of weights in the output layer of a neural network with N neurons, $\sigma(\cdot, \cdot) : \mathbb{R}^n \times \mathbb{R}^{n_u} \rightarrow \mathbb{R}^N$ denotes the vector of activation functions. The quantity $\varepsilon(\cdot, \cdot) : \mathbb{R}^n \times \mathbb{R}^{n_u} \rightarrow \mathbb{R}^n$ represents the bounded neural network approximation error.

In this proposal, we aim to develop new data-driven neuro-adaptive observer techniques which can solve the problem of the presence of several unknown parameters W , compared to the available number of system states. The use of learning techniques may reinforce the classical methods and will allow overcoming the obstacle by reducing the number of unknown parameters to consider in the adaptive observer. On the other hand, we aim to overcome the presence of some conservative and unrealistic constraints generally involved in design conditions, like equality constraints, presence of linear dependence of x and W .

– *T3.3. Combined modeling approach (RNOD&NANS):* Once T 3.1 and T 3.2 are achieved, we will tackle the strategy depicted in Figure 1(a), which consists in combining the regular nonlinear observer and the data-driven neuro-adaptive observer. Indeed, when the vehicle is connected, we have access to informations which can lead to more accurate dynamic models. Then updating the model online by using neural networks will improve the estimation and its resilience and robustness. This combined modeling architecture will lead to issues on global convergence of the combined estimation. The modeling can be viewed as hybrid system for which global stability requires some conditions like dwell-time. In this task, we will work on developing switched strategies based on vehicle assumptions.

– *T3.4. Multi-Connected Intelligent Software Sensors (MCIS):* The objective of this task consists in constructing cooperative nonlinear estimators capable to exchange informations with each other by using a connection protocol as depicted in Figure 1(b). A connection protocol means a mathematical relation between several observers/estimators/predictors. Recently, we developed a novel approach for discrete-time Linear Parameter Varying (LPV) systems

$$x_{k+1} = \left(A_0 + \sum_{i=1}^{i=n_\rho} \xi^i(\rho_k) A_i \right) x_k + Bu_k \quad (3a)$$

$$y_k = Cx_k \quad (3b)$$

capable to give an exact estimation of the states in finite-time. Indeed, it has been demonstrated in [19] that if the following matrix exists and is invertible for a given integer $m \geq 1$ and gains K_j, L_j

$$\mathbb{E}_m(k) \triangleq \left[\prod_{i=1}^m \left(\sum_{j=0}^{n_\rho} \xi_{k-i}^j (A_j - L_j C) \right) \right]^{-1} - \left[\prod_{i=1}^m \left(\sum_{j=0}^{n_\rho} \xi_{k-i}^j (A_j - K_j C) \right) \right]^{-1} \quad (4)$$

then two LPV observers with gains K_j and L_j , respectively, provide an exact and finite-time estimation of x_k for $k \geq m$. The criterion (4) is the *connection protocol* between the two observers. This task aims to generalize the idea to nonlinear systems under the form (1) and establish accordingly *smart connection protocols*. The first method we plan to follow consists in transforming the nonlinear system into an LPV one with inexact parameters. then, in addition to the connection protocol (4), appropriate optimization criteria will be necessary for robustness and performance. The use of *modulating functions*, generally used for linear systems, to enhance performances or to get finite-time convergence, is also one of the objectives of this task. In presence of a high number of nonlinearities, standard Luenberger observers may be *combined with different types of estimators/predictors*, like moving horizon estimator or other stochastic predictors, to reduce the number of unknown LPV parameters. All these ideas will be *exploited to tackle fully nonlinear models* without transformation. Also, the work in [20] is promising.

Deliverables–Risks and Fallback Solutions

Deliverables

- Open-source library of the implemented algorithms in Matlab & Python, released on github
- Graphical software interface or Toolbox for the established methodologies, posted online on the project's website

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- Publications in methodological (Automatica, IEEE TAC, SIAM-SICON, SCL) and vehicle oriented (IEEE Trans. VT, IEEE Trans. ITS) journals and in conf. proceedings (IEEE CDC, IEEE ACC, ECC, IFAC)

Risks and Fallback Solutions: WP3 is the core of the project and is the most "at risk" WP. The risk could come from overly ambitious objectives, namely the development of several rigorous methodologies that would work on the established vehicle model. To minimize the risk, the objectives are divided into several steps to guarantee the progress of the project. On the other hand, promising preliminary results for the objectives have been developed by the main contributors (with long experience in fundamental research) of this WP3 [4, 11, 19, 21, 22]. Another issue is the computational complexity due to the combined methodology based on radar measurements, computer vision, deep-learning, and the use of smart connection protocol. However, the risk related to this issue is low: IRSEEM can use the CRIANN platform (Centre Régional Informatique et d'Applications Numériques de Normandie). Also, Professor Rajamani will use powerful computational resources from his laboratory. Besides, we will also work on the development of fast optimization algorithms to reduce the computational complexity.

• **WP4: $\mathbb{R}^3$ Software Sensors for Vehicle Tracking in a Platoon**

Contributors—Deadlines—Objectives

Leader: Dalil ICHALAL (IBISC)

Partner contributors: CRAN (40p.m), IRSEEM (20p.m), IBISC (50p.m), LISEC (1p.m)

Deadlines (Start – End): T0+15 – T0+45

Objectives: This work package is devoted to resilient, robust, and reliable estimation algorithms for more complex systems than those investigated in the previous work package. The objective is to enhance more the safety of autonomous vehicles. In addition to the complexity of the nonlinearities, we will consider complex architectures of cyber-attacks, sensor faults, and data loss and develop $\mathbb{R}^3$ —software sensors covering a high number as possible of complex architectures and failures occurring in automated vehicles in a platoon. The partners have already published advanced algorithms which may be exploited and applied on autonomous vehicles. The WP4 contains 3 tasks: T4.1–T4.2–T4.3.

– *T4.1. Architectures of cyber-attacks, sensor faults, and data loss:* In this task, we will investigate several architectures of attacks and sensor faults to identify the more important issues affecting dangerously the safety of autonomous vehicles. The case of data loss will also be considered here in order to evaluate the impact on estimation. We will be particularly interested by malicious cyber-attacks on variables used in control systems. We mention, for instance, cyber-attacks under the form of injection of time-varying false data; time-varying delays; false data inside the nonlinearities. In a general way, we will study architectures under the form:

$$\dot{x} = Ax + B_1 f_{ca} + B_2 f_s + \kappa(x, f_{ca}, f_s, u) \quad (5a)$$

$$y = Cx(t - \tau(t)) + D_1 f_{ca} + D_2 f_s \text{ for } t \in \mathbb{R}^+ \setminus [\tau_1 \ \tau_2] \quad (5b)$$

where x is the system state, u is the control input, f_{ca} and f_s are the cyber-attack and sensor fault vectors, respectively. $\tau(t) \geq 0$ is a time-varying delay. The constraint $t \in \mathbb{R}^+ \setminus [\tau_1 \ \tau_2]$ represents the data loss in the interval $[\tau_1 \ \tau_2]$, which means unavailability of the measurements y in this interval. The use of a statistical predictor [2] allows evaluating the impact of data loss or to estimate it. All these variables are important for the control systems to make decisions.

– *T4.2. Software Sensors design by using neuro-adaptive UIOs:* As a first step, we aim to follow neuro-adaptive nonlinear observer design technique presented in the previous work package and improve it to deal with the design of $\mathbb{R}^3$ —software sensors, which lead to $\mathbb{R}^3$ —controllers. Depending on the cyber-attack architectures and sensor faults, the UIO should be developed accordingly. Indeed, the use of a full nonlinear

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model will lead to complex dynamics for which available standard approaches fail. The strong experience of IBISC, CRAN, and LISEC, in this field allows to handle successfully the issues related to this task. D. Ichalal and S. Mammar published an interesting approach on UIOs for linear systems [23]. Also, A. Zemouche and R. Rajamani published a joint work on resilient semi-autonomous adaptive cruise control by using an adaptive UIO [16]. These results will be generalized to more complex cyber-attacks architectures and sensor faults for nonlinear models issued from vehicle tracking problem.

– *T4.3. Software Sensors design by using multi-stage sensors and learning:* The aim of this task consists in generalizing the concept of multi-connected intelligent estimation algorithms introduced in Task 3.4. This strategy will lead to novel fault diagnostic methods and cyber-attacks detection. In addition, we will face not only fault estimation, cyber-attack detection, and data-loss recovery, but also additional internal parameters coming from the nonlinearities of the model. Indeed, our strategy here is to rewrite the nonlinearity of the system under a quasi-LPV structure with unknown parameters. Due to the number of nonlinearities, the number of unknown time-varying parameters is sufficiently large that it renders the usual necessary rank constraints not satisfied, and the approximation difficult to be achieved. Then, we aim to use learning virtual sensors to estimate such parameters.

Deliverables–Risks and Fallback Solutions

Deliverables

- Open-source library of the implemented algorithms in Matlab & Python, released on github
- Publications in journals (IJRNC, IEEE TCST, IEEE TVT, IEEE TITS) and in conf. proceedings (IEEE CDC, IEEE ACC, ECC, IFAC)

Risks and Fallback Solutions: The risks related to this WP4 are quite similar to those encountered in WP3, namely methodological aspect and computational complexity. Then the same fallback solutions are applicable due to the complementarity of the consortium. Another point may be related to the estimation of cyber-attacks and/or faults or to face data-loss. However, the risk related to this issue is low because the contributors have a solid experience in the field and have promising preliminary work [16, 23]. The issue related to modeling realistic cyber-attack, fault diagnosis, and data-loss architectures encountered in connected vehicles can be overcome by our partners FAAR and LISEC.

• **WP5: Validation, Tests, and Integration of Algorithms**

Contributors–Deadlines–Objectives

Leader: Randolph TOOM (FAAR)

Contributors: CRAN (20.8p.m), IRSEEM (30p.m), IBISC (30p.m), FAAR (10p.m), LISEC (1p.m)

Deadlines (Start – End): T0+12 – T0+48

Objectives: All the methods developed in the previous work packages will be integrated and implemented, and they will be applied and validated on systems acquired by the research team (FAAR, IRSEEM, IBISC) as well as new platforms of scaled vehicles to be acquired (CRAN). The validation task will also consider implementation of the estimation algorithms on some control systems related to automated vehicles in a platoon, namely: following one of the target vehicles, changing lanes in order to avoid a stalled vehicle in the same lane. The goal here will consist in switching from the standard controller to another one when cyber-attacks or faults are detected. This controller will use the $\mathbb{R}^3$ software sensors established in the previous work packages. The multidisciplinary of partners is essentially very benefit for this work package. Indeed, the 5 partners will work together and exchange their experience on vehicle modeling, instrumentation and integration, deep-learning and computer vision, and nonlinear estimation. The WP5 contains 3 tasks: T5.1–T5.2–T5.3.

– *T5.1. Validation in 3D simulation environments:* All the algorithms developed during the project will be first tested and validated in simulation using high fidelity simulators and virtual reality environments as the one developed by LISEC (**AutoControl**). 3D simulators are also available from FAAR and IRSEEM. A CARSIM software is requested in this project by CRAN in the goal to facilitate testing algorithms by PhD

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and Master students. Simulations with experimental data from LISEC will also be performed.

– *T5.2. Validation in scaled and real vehicles in laboratory:* After the algorithms have been satisfactory tested and validated in simulation, they will be tested in a new experimental platform of scaled vehicles (requested in the project by CRAN) and in instrumented real vehicles in laboratory (Renault Espace from IRSEEM; Renault ZOE from IBISC). The Autonomous Navigation Laboratory of IRSEEM will be put under the disposal of the project for the purpose of validating the estimation algorithms developed in the project. The IRSEEM experimental platform (735€/day) proposes a complete service including materials, automotive equipments, technicians and engineers according to the needs of the project.

– *T5.3. Integration and Tests with real vehicles in FAAR SAS facilities:* After the algorithms are validated in laboratories, they will be integrated by FAAR in advanced real platform of instrumented vehicles.

Self-e-Car® is the innovation platform, of FAAR, open to the development of autonomous and connected vehicle technologies intended to accommodate its technological bricks and those of its partners (large groups, startups). This powerful tool allows testing and improving environmental perception technologies, intersection management and vehicle control. Hence the complementarity of FAAR with the other partners of the ArtISMo project. The provision of the platform for testing is 1k€/day, and it is available for sale at 450k€. This platform will be made available to the ArtISMo project free of charge, which is a precious advantage for the success of the project.



Figure 3: Self-e-Car® FAAR platform

Deliverables–Risks and Fallback Solutions

Deliverables

- Open-source library on the validation results, including CARSIM simulations and codes
- Tutorial on the experimental platform with scale vehicles (instrumentation, algorithms, and results)
- Final software sensors system for vehicle tracking as handled in ArtISMo project
- Implementation of the algorithms in the FAAR electronic communication calculator (TCU), which will be equipped with 5G and LiFi
- Publication of a survey paper in IEEE Control Systems Magazine or in Technology journal

Risks and Fallback Solutions: In general, the critical risks related to integration and testing steps are the unavailability of realistic instrumented platforms and lack of human resources (Engineers). However, in our case, the consortium involves the industrial company FAAR, which puts at the disposal of ArtISMo project: 2 instrumented vehicles; 4 engineers expert in instrumentation, testing, and validation. Experimental platforms are also available from the partners IBISC and IRSEEM. The engineers will be closely supported by the contributors of the estimation theory to adapt the methodological algorithms and improve them to avoid any risk. The risk related to the computational complexity is low since FAAR puts enough computer science resources at the disposal of the project. FAAR has contact with manufacturers in order to update its range of products. Today several of these manufacturers already allow neural networks to run on ECU thanks to the latest processors (nvidia, Horizon, Baidu, NXP, ST, Kalray). This will migrate the software developed within this project to these platforms.

2 Organisation and Implementation of the Project

2.1 Scientific coordinator and its consortium / its team

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2.1.1 Scientific coordinator and his experience

Ali ZEMOUCHE, from CRAN, is experienced in nonlinear observer design and estimation with application to autonomous vehicles. He will be involved in all aspects of the project to ensure the successful delivery of the project. He will provide the overall coordination and leadership of the program. He will specifically focus on the development of intelligent neuro-adaptive estimation algorithms for connected vehicle applications (vehicle tracking, cyber-attacks detection, estimation of unexpected obstacles, etc). He will also be involved in the validation and integration of the developed algorithms, on simulation/experimental platforms (CARSIM, instrumented vehicles). He is involved in scientific projects and published papers (related to the proposal) in top international journals like [3, 24, 21, 22]. He is very well cited in the literature and his papers have become famous in the field of nonlinear estimation. He is invited by many universities to give plenary presentations on nonlinear observers. Ali ZEMOUCHE is involved and works on the following past and current projects:

1. **PHC–SIAM 42844ZK (2019–2020, coordinator):** *Observers for vehicle tripped rollover prevention.*
2. **Contract with SEGULA Technologies (03/2020–08/2020, coordinator):** *Vehicle tracking, observers.*
3. **Partnership with University of Minnesota (2017–2020):** *Autonomous and connected vehicle.*
4. **ANR GLMAV (2010–2013, involved - supervision of a Postdoc):** *Gun Launched Micro Aerial Vehicle.*
5. **DGA project (2007–2010, involved - PhD supervision):** *Position and attitude estimation of a projectile.*

As project coordinator, he will lead the steering committee of the ArtISMo project to ensure the evolution of the different tasks and the synergy between the partners, monitor planning and ensure deadlines for the completion of deliverables, and organize meetings.. Since nonlinear observer design and estimation are the main research topics of the project coordinator, he will secure **50%** of his research activities to work actively on all the WPs of this project.

2.1.2 Presentation of the consortium and its complementarity

Presentation of the consortium

1. CRAN: A. Zemouche (*coordinator*), M. Zasadzinski, C. Delattre, H. Rafaralahy, M. Alma, L. Boutat-Baddas.

CRAN (Research Center for Automatic Control of Nancy, UMR 7039) is a research laboratory with high experience on automatic control theory and its applications. In particular, the members from CRAN involved in this project are internationally recognized in the field of nonlinear estimation and observer design with application on several real-world models, namely UAV, projectiles, biogas processes, robotics, and autonomous vehicles. The team has been and is involved in numerous projects; we can mention the recent projects like DGA-REI 2008/2011, ANR-09-SECU-12-GLMAV, ANR Emergence-11-EMMA-035 PROTHERMOVERRE, PHC SIAM 42844ZK. For five years, the team have worked closely with University of Minnesota on project on autonomous and connected vehicles. Hence, the team covers a large number scientific topics related to the proposal, namely nonlinear observer design, adaptive observers, stochastic systems, unknown input estimation, optimization, and application to autonomous and connected vehicles. CRAN is the coordinator of the project and the leader of both WP1 and WP3.

2. IRSEEM: S. Ahmed Ali (*leader*), R. Khemmar, A. Cabani, Y. Dupuis, R. Hajri El Houda, N. Langlois, D. Quangtruc. IRSEEM team conducts a research partnership, with an application purpose. The industrial sectors of the automobile, aeronautics, renewable energies, telecommunications and healthcare are thus targeted. The IRSEEM laboratory conducts research aiming at bringing new embedded smart functionalities for intelligent transportation systems and mobile robotics in order to make them autonomous and cooperating. Research work related to embedded vision and multi-sensor perception for driving assistance and autonomous navigation have been conducted for several years. IRSEEM has conducted several research projects as a lead partner such as: ADAPT, SAVEMORE, VIRTUOSE, ARGOS, etc. IRSEEM is the leader of WP2.

3. IBISC: D. Ichalal (*leader*), S. Mammar, L. Nouvelière, K. Lahiani.

The IBISC laboratory (Informatique, BioInformatique, Systèmes Complexes EA 4526) is a laboratory of the University of Evry Paris-Saclay which is structured into 4 research teams: AROBAS, COSMO, IRA2 and SIAM. The members enrolled in the project are part of the SIAM team (Signal, Image, AutoMatique). This team is developing automatic methods (Control, observation, diagnosis) but also signal, image and perception processing based on artificial intelligence approaches. These methods are developed for applications in the transport, vehicles and biomedical fields. The members involved in this project are recognized for work on

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driver assistance systems using automatic control tools. IBISC is the leader of WP4.

4. LISEC–Minnesota: Rajesh Rajamani (leader).

The research in LISEC laboratory focuses on the design of devices, estimation algorithms and controllers for smart and autonomous systems. Specific major areas of research in LISEC lab have included: Estimation algorithms and control systems for intelligent vehicles, active sensing and estimation on smart bicycles, imminent collision detection sensors, tracking systems for complex traffic on urban roads and highways, and road condition monitoring transducers. The head of the lab is Professor Rajesh Rajamani, an international leader in vehicle dynamics and control. He is author of 16 patents and he got more than 50 significant grants. He is the co-founder of the start-up *Innotronics* based on technologies developed in LISEC.

5. FAAR SAS: R. Toom (leader), E. Salvator, E. Nang-Tchee, M. Duthoit, T. Marchand.

FAAR is specialized in the development of electronic embedded systems, from prototyping to production programs. FAAR SAS is a great company with a turnover of **8 M€**. FAAR works mainly on the development of new technologies for smart mobility. The main partners of FAAR are automotive companies and industrials like Renault, PSA, Safran, Thales, Alstom, nVidia. FAAR has a long experience in commercialization of new technologies related to autonomous vehicle, namely SELF-e-CAR, CROSS-DRIVE, EASY CAN, and several electronic control units and configurable displays. FAAR has the expertise, the skills and the partners to go beyond the state of the art and respond to the objectives of smart mobility addressed in this project. FAAR is the leader of WP5 and offers a high technology vehicle platform (*details in T5.3*). Today, FAAR is involved in various autonomous driving projects: vehicle automation, design of the electronic and electrical architecture, and development of specific computers monitoring an autonomous vehicle.

Complementarity of the consortium

The proposal revolves around four main points: *modeling, benchmarking, deep-learning, computer vision; estimation theory for nonlinear systems; fault diagnostic, vehicle dynamics, and embedded systems; connectivity, companies transfer, and commercialization*. The makeup and balance of the consortium is essential for the successful delivery of the project as the consortium covers all important aspects of estimation in intelligent transportation systems and allows several inter-related work-packages running in parallel thus achieving maximum possible outcomes within the specified timeframe. Each scientist involved in the project is expert in a field, covering the whole project from the development of comprehensive theory to the integration of the algorithms on instrumented vehicles and commercialization of the product. The five partners complement each other. The company “FAAR” will bring its expertise in integrating new technologies for autonomously running transport (Control systems, automation, connectivity, electronic communication calculator (TCU)) and its *strategy in commercialization of products for industrial partners and major automotive groups like PSA, Renault, Safran, Thales, Alstom, nVidia*. “IBISC” has a long expertise in *fault diagnostic and control design for vehicle applications*; “IRSEEM” has an *autonomous navigation platform* and has a strong background in *modeling and benchmarking of several applications, deep-learning and environment perception*, namely for autonomous navigation within *embedded computer vision algorithms*. “CRAN” has a *solid experience in nonlinear estimation from theoretical and real-world application (autonomous vehicles)* viewpoints. Finally, Professor R. RAJAMANI, from “LISEC–University of Minnesota”, who is *internationally recognized in vehicle control dynamics*, will participate to the implementation of new estimation algorithms and *will strengthen the project in innovations related to vehicle dynamics and control*. He will share his highly recognized experience. A research project as multidisciplinary as ArtISMo, can only succeed if it combines competences of these five partners. Such a complementarity may be represented as in Figure 4, which shows connections between the 5 partners (thin arrows) and the main expertise of each partner (thick arrows). Our industrial partner, FAAR, is a main key of complementarity.

Requested human resources for the project

To reach the all scientific objectives of the ArtISMo project, we would like to hire: 2 PhD students, 2 Postdoc researchers, and 12 Master interns. The PhD1 will work on the design of learning-based neuro-adaptive observers for vehicle tracking (WP3), under the co-supervision of Ali ZEMOUCHE & Cédric DELATTRE (CRAN) and

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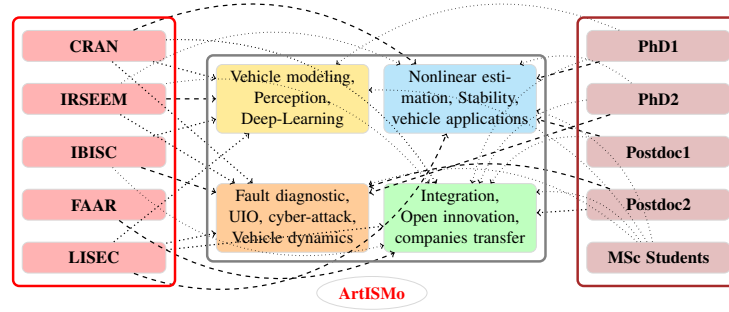


Figure 4: Complementarity of the consortium in ArtISM0 project covering all the WPs.

Redouane KHEMMAR (IRSEEM). The PhD2 will work on the design of $\mathbb{R}^3$ -software sensors by considering faults, complex cyber-attack architectures, and data-loss (WP4), under the co-supervision of Dalil ICHALAL & Said MAMMAR (IBISC) and Marouane ALMA (CRAN). The PostDoc1 will work on the task T3.4 to strengthen and complete the objectives of the WP3 and WP5 and develop multi-connected intelligent software sensors by integrating stochastic predictors and optimization-based estimators. The PostDoc2 will work on complex cyber-attacks architectures by considering virtual sensors and learning. Both PostDoc1 and PostDoc2 will work in collaboration with the all partners. As for MSc students, some of them will work with PhD1, PhD2, PostDoc1, and PostDoc2. The other MSc students will strengthen the instrumentation and validation tasks.

Collaborations related to the proposal's topic

This project will take advantage of ongoing collaborations with international partners:

- Ankush CHAKRABARTY and Mouhacine BENOSMAN from Mitsubishi Electric Research Laboratories, USA, working on learning-based observers.
- Gridsada PHANOMCHOENG from *smart mobility research center*, University of Chulalongkorn, Thailand, who is involved in a common PHC–SIAM grant for 2 years (2019/2020), with Ali ZEMOUCHE on observer design for vehicle tripped rollover prevention.

These collaborations will strengthen the project through commun discussions on the topic and will offer opportunities for the recruited PhD students and PostDoc researchers to have scientific stays. In order to support trips, we will use PHC projects and funding provided by our labs.

2.1.3 Implication of the scientific coordinator and partner's scientific leader in on-going projects

The following Table 3 includes informations concerning the involvement of the scientific coordinator and partner's scientific leader in regional, national and international on-going projects:

Name of the researcher	P.M	Call, funding agency, grant allocated	Project's title	Name of the coordinator	Start–End
Ali Zemouche	10	PHC, Campus France, 15 300€	SIAM–42844ZK	Ali Zemouche	01/2019–12/2020
Sofiane Ahmed Ali	3.8	Normandy region, 2216.4 K €	THESIS project	Romain Dupperier	11/2017–09/2020
Dalil Ichalal	×	×	×	×	×
Randolph Toom	×	×	×	×	×
Rajesh Rajamani	4	NSF:BIC– \$999 731	PFI–1631133	Rajesh Rajamani	09/2016–08/2020
	3	NIH–\$508 000	BRAIN Initiative	Rajesh Rajamani	09/2016–06/2021
	2.5	NSF, EFRI C3 SoRo \$ 550 189	EFRI–18309585	Rajesh Rajamani	08/2018–07/2022

Table 3: Implication of the scientific coordinator and partner's scientific leader in on-going projects. The symbol “×” means that the corresponding scientific leaders have no ongoing projects. Their projects were completed this year.

2.2 Implemented and requested resources to reach the objectives

Table 4 summarizes the global requested funds in accordance with the information filled out on the website and with ANR's grant allocation rules. More details on the staff expenses, travel costs are given in Table 5.

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		Partner 1 CRAN	Partner 2 IRSEEM	Partner 3 IBISC	Partner 4 FAAR SAS	Partner 5 LISEC
Staff expenses		166 700,16€	180 923 × 50% = 90 461,50€	110 702,88€	42500 × 45% = 19 125€	0.00€
Instruments and material costs (including the scientific consumables)		7 000€	5000 × 50% = 2 500€	5 000€	15000 × 45% = 6 750€	0.00€
Building and ground costs		0.00€	0.00€	0.00€	0.00€	0.00€
Outsourcing / subcontracting		12 000€	0.00€	2 000€	0.00€	0.00€
General and Admin. costs & other operating expenses	Travel costs	30 000€	10 000 × 50% = 5 000€	12 000€	16000 × 45% = 7 200€	0.00€
	Administrative management & structure costs	17 256,01€	124078 × 50% = 62 039€	10376,23€	31070 × 45% = 13 981,50€	0.00€
Sub-total		232 956,17€	160 000,32€	140 079.11€	47 056.50€	0.00€
Requested funding		580 092.10€				

Table 4: Requested means by item of expenditure and by partner.

	CRAN	IRSEEM	IBISC	FAAR	LISEC
<i>Nb of permanent researchers</i>	6	6	4	5	1
<i>Nb of non-permanent researchers</i>	-	1	-	-	-
Staff expenses:					
— PhD grant	92	-	96	-	-
— PostDoc grant	60.3	41.25	-	-	-
— Internship grant	14.4	8.4	14.4	-	-
— Permanent staff expenses for full cost structures (Requested: 50% for IRSEEM, 45% for FAAR)	-	40.812	-	19.125	-
Sub-total	166.7	90.462	110.4	19.125	-
Travel costs:					
— Meetings	4	1	4	3.6	-
— Meeting CRAN/LISEC - Minneapolis	2	-	-	-	-
— Organization of Workshop	8	-	-	-	-
— International Conferences	16	4	8	3.6	-
Sub-total	30.00	5.00	12.00	7.2	0.00

Table 5: Details on staff expenses and travel costs in k€. The symbol “–” means that there is no cost requested by the corresponding partner / no non-permanent researchers involved in the corresponding partner.

Staff expenses - CRAN and IBISC will each hire a PhD student (36 p.m), each working on a specific WP. To ensure collaboration between partners, the PhD will be co-supervised by two different partners. The CRAN and IRSEEM will each hire a PostDoc (18 p.m):

- PhD1 at CRAN will work on WP2/WP3/WP5, under the joint supervision CRAN and IRSEEM.
- PhD2 at IBISC will work on WP2/WP4/WP5, under the joint supervision IBISC and CRAN.
- PostDoc1 (18 months) at CRAN will work on WP3/WP5, more specifically on T3.4, T5.1 and T5.2.
- PostDoc2 (18 months) at IRSEEM will work on WP4/WP5 in collaboration with IBISC and FAAR.
- 12 MSc interns will be hired (12 × 6 p.m): the CRAN, IRSEEM, and IBISC will hire each 4 interns (1 intern per year). The MSc students will work on the instrumentation task (T5.2); and to investigate performances of estimation algorithms by real-time simulations under 3D simulation environments (CARSIM,... etc). They will participate and help on the four workpackages WP2–WP5.
- The two partners IRSEEM and FAAR are full cost structures. Then staff expenses are included in the requested budget: 6 and 5 permanents for IRSEEM and FAAR, respectively, as depicted in Table 4 and Table 5. The IRSEEM and FAAR request respectively 50% and 45% of the budget.

Travel expenses - The travel expenses will cover:

- *plenary meetings at Longwy, Rouen, or Evry, where the partners are located:* 3 per year for 8 people from CRAN, 5 people from IBISC, 3 people from IRSEEM and 5 people from FAAR.
- *meetings for open innovation & companies transfer as explained in T1.3:* FAAR will manage this task and requests 1.8 k€.
- *meeting at LISEC in Minneapolis:* between the coordinator Ali ZEMOUCHE and Rajesh RAJAMANI.

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- *organization of an international workshop at CRAN (8 k€):* A two-days Workshop will be organized at the end of the project. We plan to invite 4 international experts (4 k€ for travel costs, 2.4 k€ for accommodation, 1.6 k€ for catering and coffee breaks) $\approx$ 30 participants.
- *participation to international conferences with an estimated cost of 2 k€ per mission:* 2 missions per year for CRAN; 1 mission per year for IRSEEM, IBISC, and FAAR, respectively.

Instruments and material costs

- We require a workstation for each PhD and the 2 postdoctoral students with an estimated cost of 2 k€ each.
- Each partner requires a workstation for each experimental platform of 2 k€ each.
- The CRAN requires $2 \times 1/8$ scale vehicle & their instrumentation (Cables, connectors, etc) for real-time validation (**WP5**): 2 k€. The IRSEEM requires instrumentation material for the autonomous navigation platform: 2 k€. The IBISC requests 3 k€ for its instrumented vehicle (Cables, connectors, electronics, radars). FAAR company requests 6.75 k€ for Sensors (LIDAR, RADAR, Long range RADAR, and Cameras) and for the instrumentation of the autonomous navigation platform (Cables, connectors, electronics boards, etc).

Outsourcing / subcontracting

- The CRAN requests a CARSIM software licence for real-time simulation: 10 k€ $\rightsquigarrow$ for task **T5.3**.
- The IBISC and CRAN request fees of 2 k€ for each, for extra pages and open access publications.

3 Impact and Benefits of the Project

3.1 Scientific impact

This project will make the following fundamental contributions to the development of estimation algorithms for smart autonomous systems:

- The project will develop estimation algorithms for nonlinear dynamic systems without resorting to linearization.
- The project will enable simultaneous online updates of both plant model parameters and state variables by utilizing neuro-adaptive observers. The neuro-adaptive observers will be developed in a rigorous analytical framework with global stability guarantees.
- The project will enable estimation in connected systems, considering a number of network architectures that include both nearest neighbor and centralized leader type of connections.
- The project will enable automated monitoring of cyber-attacks and fault occurrences in a connected nonlinear systems architecture.

All of the above fundamental contributions will have a significant impact on enabling new functions and performance in autonomous systems. As automation, instrumentation, feedback and autonomous monitoring continue to increase everyday all around us, these fundamental contributions to estimation algorithms have the potential to significantly play an increasingly important role in the modern world. In terms of the scientific impact on the world of autonomous and connected vehicles, the developed algorithms have a very real potential of making autonomous transportation viable, of making autonomous vehicles reliable through monitoring of cyber-attacks and sensor/actuator faults, of increasing traffic capacity of roads through enabling closer spacing between vehicles for higher traffic capacity, and of increasing safety by assisting humans to ensure a much lower accident rate compared to today where over 90% of accidents are caused by human error.

3.2 Actions to transfer technology and innovation to the social and economic world

Several of the algorithms developed in this project can lead to commercializable products. For instance, development of estimation algorithms that predict trajectories of other downstream vehicles on the highway have a large number of applications beyond autonomous vehicles. Such algorithms can be used to predict the evolution of traffic on roads, to predict the impending developing traffic jams and to potentially disrupt them, to provide accurate future velocity estimates to hybrid electric vehicles in order to improve energy management and fuel economy, and to improve performance of safety-assist systems such as adaptive cruise control and lane-keeping.

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Thus a device that provides estimated prediction of downstream vehicle trajectories can be commercialized in the market place due to its significant value in a broad range of applications. **The involvement of FAAR company in this project will allow achievement of this commercialization task. The experience of FAAR in commercialization of new technologies related to vehicle (for instance: SELF-e-CAR, CROSS-DRIVE, EASY CAN, and several electronic control units and configurable displays) will be of great help.** We will also engage with professional bodies and with the ANR team to identify other opportunities for exploitation by considering ways to transfer this innovation to companies collaborating with our organizations and specifically through our industrial partner, FAAR, willing that their potential customers they are currently in contact with will integrate the outcome of this project for their products. FAAR considers that only high quality innovation solution-based on artificial intelligence can enable commercialization of autonomous vehicles on a large scale.

In addition to near-term technology spin-offs such as the one described above, the project will also have a significant potential to transfer technology to large automotive companies through licensing of specific algorithms that are likely to be valuable to companies that work on autonomous vehicle technologies. **FAAR partner has commercialized, in the past, products and prototypes for several automotive companies and industrial partners like Renault, PSA, Safran, Thales, Alstom, nVidia.**

It is worthy to notice that Professor Rajesh RAJAMANI from University of Minnesota has a long experience in commercialization of products and innovation in writing several patents related to vehicle dynamics. He is the co-founder of the start-up company "*Innotronics*" based on technologies developed in his lab¹. Also, notice that the consortium is involved in 20 patents and is very close to the start-up company *TechnOptiz*.

Enhancing the safe control and operation of autonomous and connected vehicles will have a great impact in many societal aspects. Achieving safe control and operation of autonomous vehicles has great benefits, such as minimizing traffic accidents by reducing the burden on humans in the driving task, improving the lives of citizens with low physical mobility by providing them door-to-door transportation services, reducing congestion via route sharing, decreasing energy consumption and pollution using electric vehicles. Hence, the improvements introduced in this project will not only have a profound impact on human health, but also will have many other great social and economic benefits.

3.3 Dissemination and valorization of the results

This project plans to intensively disseminate results while also taking regular steps to protect intellectual property and transfer technology to commercial organizations wherever possible. The dissemination of research results will follow along the following steps:

1. Consideration of whether a particular algorithm and estimation system needs to be protected through filing of patents or filing of copyright.
2. Submission of papers to reputed European and international conferences, after having filed for patent applications in the cases where this is necessary. Publication in proceedings of renowned conferences together with in-person presentation of results at these conferences.
3. Publication of enhanced and complete results through high quality archival journal papers.
4. Copyrighting of estimation algorithms, encryption of software code as needed, and dissemination of software code to the worldwide community. The objective here is to make the code available to the community that will find it useful while also trying to enhance the reputation of the research group that developed the software.

In order to improve the quality of the results dissemination (publications, reports, tutorials, etc.) which will be developed through the project and to offer an open access for not only project partners, peer-reviewed but also readers and end-users, we will make within the first work-package the following strategy: 1) sharing publications within an open archive called HAL², where authors can submit research documents from all academic fields; 2) ESIGELEC partner has an online open source platform called forge³ which will be used to share not only all publications, but also technical reports, software development, tutorials, etc. 3) We will use also a project common page within Github.

1. <https://ncet2.org/component/sppagebuilder/?view=page&id=51>
2. <https://hal.archives-ouvertes.fr/>
3. <https://forge.esigelec.fr/>

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